

Math 411 Summer 2025**Midterm 1****July 23, 2025****8:30 - 9:30 AM**

Name: _____

Student ID: _____

Instructions:

1. You have 60 minutes to finish the exam.
2. You may only work on your own.
3. Please try to write as legible as you can.
4. Do not write on the back of any page.
5. Please provide as many detail as you can.
6. Write down and sign your honor statement.

Question:	1	2	3	4	Total
Points:	6	8	12	24	50
Score:					

1. Let G be a group and R be a commutative ring. Write down the following.
 - (a) (2 points) The statement of the division algorithm.
 - (b) (2 points) The order of the group element $g \in G$ is ...
 - (c) (2 points) An element $r \in R$ is irreducible if ...

2. Find all integer solutions to the following equations.

(a) (4 points) $7x \equiv 4 \pmod{23}$

(b) (4 points) $x^5 - x + 2 = 0$

Proof. If $x^5 - x + 2 = 0$ has solution in $\mathbb{Z}$, then $x^5 - x + 2 = 0$ has solution in $\mathbb{Z}_5$. We show $x^5 - x + 2 = 0$ has no solution in $\mathbb{Z}_5$. First, note $x = 0$ in $\mathbb{Z}_p$ is not a solution. Recall by Fermat's little theorem, $x^5 \equiv x \pmod{5}$ for $x \not\equiv 0 \pmod{5}$. It follows that $x^5 - x \equiv 0$. Therefore $x^5 - x + 2 \equiv 2 \not\equiv 0 \pmod{5}$. $\square$

3. Recall $\binom{p}{k} = \frac{p!}{k!(p-k)!}$ for integers $0 \leq k \leq p$. Determine if the following statement is true or false by proving/explaining it or giving an (counter)example.
- (a) (4 points) Let $n > 4$ be integer. Can $(n-1)! \equiv 0 \pmod n$ ever hold?
 - (b) (4 points) Let $p > 0$ be a prime number in $\mathbb{Z}$. Assume $N = \binom{p}{k}$ is an integer. Is it always true that $p \mid \binom{p}{k}$ for $0 < k < p$?
 - (c) (4 points) Let $p > 0$ be a prime number in $\mathbb{Z}$. Is $(x+y)^p \equiv x+y \pmod p$ always true for all integers x, y ?

Proof. 1. Yes. Take $n = 6$.

2. Yes. Note $N = \frac{p!}{k!(p-k)!}$ implies $N \cdot k! \cdot (p-k)! = p!$. In particular $p \mid N \cdot k! \cdot (p-k)!$. Clearly $p \nmid k!$ and $p \nmid (p-k)!$ for $0 < k < p$. By Euclid's Lemma, $p \mid N$.

3. Yes. Note that $(x+y)^p = x^p + \sum_{0 < k < p} \binom{p}{k} x^k y^{p-k} + y^p$. By part b, $p \mid \binom{p}{k}$ which implies $(x+y)^p \equiv x^p + y^p$. Since p is prime, by Fermat's little theorem, $x^p + y^p \equiv x + y$ when both x, y nonzero. When either x or y is zero, and when both $x, y = 0$, the equality can be easily checked. $\square$

4. Consider the domain $\mathbb{Z}[\sqrt{-2}] = \{a + b\sqrt{-2} : a, b \in \mathbb{Z}\}$. Recall $\sqrt{-2} = \sqrt{2}i$ where $i^2 = -1$.

(a) (4 points) Write down (and show) a multiplicative norm on $\mathbb{Z}[\sqrt{-2}]$.

Define $N(a + b\sqrt{-2}) = a^2 + 2b^2$.

(b) (4 points) Determine and explain if $\mathbb{Z}[\sqrt{-2}]$ is a field.

Note if $a + b\sqrt{-2}$ is a unit, then $a^2 + 2b^2 = N(a + b\sqrt{-2}) \mid 1$ since the norm is multiplicative.

This is not a field because 2 has norm 4 which can't be a unit. In particular, $a + b\sqrt{-2}$ is a unit iff it is ± 1 .

(c) (6 points) Determine and explain if 3 is an irreducible element in $\mathbb{Z}[\sqrt{-2}]$.

First note if $a + b\sqrt{-2}$ has prime norm, then it is irreducible because if $a + b\sqrt{-2} = xy$, then $N(a + b\sqrt{-2}) = N(x)N(y)$ as the norm is multiplicative and, by part a, the units are precisely ± 1 . No because $3 = (1 + \sqrt{-2})(1 - \sqrt{-2})$. Note that $1 \pm \sqrt{-2}$ has norm 3 which is a prime in $\mathbb{Z}$. So both are irreducible. This implies 3 is not irreducible.

(d) (6 points) Write down and explain an example of a maximal ideal in $\mathbb{Z}[\sqrt{-2}]$.

This is an Euclidean domain. So irreducible elements generates maximal ideals. In particular, $(1 + \sqrt{-2})$ is a maximal ideal as we showed it is irreducible in part c.

(e) (4 points) Determine and explain if 3 a prime element in $\mathbb{Z}[\sqrt{-2}]$?

No. If it is prime, then it is irreducible. But by part b, 3 is reducible.

Scratch paper. If you write any answer here, make sure you label the question clearly.

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